

MEMORANDUM: THE LOCAL FACTORS BEHIND THE TWIST-SYMMETRISED BOUND

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ABSTRACT. This memorandum answers the question raised by both referee reports on the note *Local component groups and the ℓ -adic valuations of the modular degree* (since restructured and retitled *Local factors of the adjoint L -function and the ℓ -adic valuations of the modular degree*; the numbering below refers to the restructured version): are the twist-symmetrised local weights of its former Conjecture 1.3, now the inequality (1.3) of its Proposition 1.3, the local factors of the adjoint / congruence-ideal theory, and if not, what are the exact local factors? The answer is that the local term of a bad prime $q \neq \ell$ is the ℓ -adic valuation of the Euler factor of the adjoint L -function at $s = 1$ that the *naïve* adjoint L -function of Diamond, Flach and Guo omits, plus the Tamagawa exponent of the adjoint representation when the reduction is potentially multiplicative. The weights of the note were its shadow. The new rule is sharper and holds with no exception at $\ell = 3, 5, 7, 11, 13$ on every optimal curve of conductor at most 500000 and on 893 curves outside the tables (with PARI's symmetric-square Euler factors at every bad prime, the wild primes 2 and 3 included). The Euler term of a prime of type III can be added as well except, at $\ell = 3$ only, when it meets a Tamagawa term at a prime $\equiv -1 \pmod{3}$: a configuration of a precise shape, which is exactly the one excluded by the hypotheses of Kim and Ota. Two parts of the rule are proved here: the Euler terms of the types I_0^* and I_n^* , with no hypothesis beyond the irreducibility of $E[\ell]$ and $\ell \nmid 2N$ (from a formula of Diamond–Flach–Guo and an elementary twisting argument), and the Tamagawa terms for $\ell \geq 5$ (from Kim–Ota). The rest is reduced to a statement about the Σ -imprimitive adjoint Selmer group that the same theory controls.

1. SUMMARY

1.1. Notation. $E/\mathbb{Q}$ is the $X_0(N)$ -optimal curve of a non-CM isogeny class, $f = f_E$ its newform, ℓ an odd prime with $E[\ell]$ irreducible, r_E the congruence number, so that $v_\ell(\deg \varphi_E) = v_\ell(r_E)$ when $\ell^2 \nmid N$ (Agashe–Ribet–Stein). $V = V_\ell E$, $\rho = \rho_{E,\ell}$, $\bar{\rho}$ its reduction, $A_f = \text{ad}^0 V$ the adjoint. We follow the conventions of Diamond, Flach and Guo [2, §1.7.2]: for a prime p let $\delta_p = \dim V^{I_p}$, so $\delta_p = 2$ for good reduction, 1 for multiplicative reduction, 0 for additive reduction; the local factor $L_p(A_f, s)$ of the adjoint L -function is the true one, and the *naïve* factor is $L_p^{\text{nv}}(A_f, s) = L_p(A_f, s)$ if $\delta_p > 0$ and 1 if $\delta_p = 0$, which is what the Euler product of $\sum a_n(f)^2 n^{-s}$ produces. The *exceptional* primes are those with $\delta_p = 0$ and $L_p(A_f, s) \neq 1$; in the normalisation of the symmetric square, $L_p(\text{Sym}^2 f, s) = L_p(A_f, s-1)$, and PARI's `lfunsympow(E,2)` returns these factors at every prime, including 2 and 3.

Definition 1.1. For a bad prime $q \neq \ell$ put

$$e_\ell(E, q) = v_\ell \left(\frac{L_q(A_f, 1)^{-1}}{L_q^{\text{nv}}(A_f, 1)^{-1}} \right) = v_\ell \left(\frac{P_q(q^{-2})}{1 - a_q(E)^2 q^{-2}} \right), \quad t_\ell(E, q) = \begin{cases} v_\ell(-v_q(j_E)) & \text{if } v_q(j_E) < 0, \\ 0 & \text{otherwise,} \end{cases}$$

where $P_q(X)$ is the polynomial with $L_q(\text{Sym}^2 f, s) = P_q(q^{-s})^{-1}$. So e_ℓ is the valuation of the Euler factor that the naïve L -function omits (it is 0 at multiplicative primes, where the naïve factor is the true one), and t_ℓ is the Tamagawa exponent of $\text{ad}^0 V$ at a potentially multiplicative prime, the ℓ -part of the valuation of the Tate parameter, which is the index n of the Kodaira

symbol I_n or I_n^* at odd q (at $q = 2$ a symbol I_n^* can have potentially good reduction and then $t_\ell = 0$). Both are invariant under quadratic twist.

For tame primes $q \geq 5$ the Euler term is read off the Kodaira type (Section 3): with χ_{-3}, χ_{-4} the quadratic characters of $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(i)$, E' the twist of E by q^* , and α, β the roots of $X^2 - a_q(E')X + q$ (type I_0^* only),

type at $q \geq 5$	$L_q(A_f, s)^{-1}$	$e_\ell(E, q)$
I_n	$1 - q^{-1-s}$	0
I_n^*	$1 - q^{-1-s}$	$v_\ell(q^2 - 1)$
II, II^*, IV, IV^*	$1 - \chi_{-3}(q)q^{-s}$	$v_\ell(q - \chi_{-3}(q))$
III, III^*	$1 - \chi_{-4}(q)q^{-s}$	$v_\ell(q - \chi_{-4}(q))$
I_0^*	$(1 - \frac{\alpha}{\beta}q^{-s})(1 - q^{-s})(1 - \frac{\beta}{\alpha}q^{-s})$	$v_\ell(q - 1) + v_\ell((q + 1)^2 - a_q(E')^2)$

1.2. The rule.

Conjecture 1.2 (Euler-factor rule). *Let E be optimal, non-CM, ℓ odd, $E[\ell]$ irreducible. Then*

$$(1) \quad v_\ell(\deg \varphi_E) \geq \sum_{\substack{q|N, q \neq \ell \\ \text{type} \neq III, III^*}} e_\ell(E, q) + \sum_{q|N} t_\ell(E, q) + [\ell = 3] w_3(E),$$

where $w_3(E)$ is the weight of the prime 3 itself when $3 \mid N$ and the reduction at 3 is potentially good: 1 for types II, IV, III^* , 2 for II^*, IV^* , 0 for III, I_0^* (an empirical rule; see Section 7). The Euler term of a prime of type III or III^* can be added as well except possibly when some prime $p \equiv -1 \pmod{\ell}$ has $t_\ell(E, p) \geq 1$.

Every term is attained as a minimum, in isolation, over tens of thousands of curves (Table 1); (1) holds with no exception on the four ranges at $\ell = 3, 5, 7, 11, 13$ (Table 2); with the type III terms included it has exceptions only at $\ell = 3$ (Section 7.3). It implies the two former conjectures of the note, its inequalities (1.2) and (1.3), because $3 \mid q - \chi_{-3}(q)$ and $3 \mid q^2 - 1$ for every $q \geq 5$.

1.3. What is proved. Theorem 4.1 (twist monotonicity of the congruence number), Theorem 5.1 (the Euler terms of the types I_0^* and I_n^* , under $\ell \nmid 2N$ and $E[\ell]$ irreducible only), Theorem 5.2 (the Tamagawa terms, $\ell \geq 5$, from Kim–Ota) and their combination, Corollary 5.3. For the types II, IV, III and their duals the rule is reduced in Section 6 to a statement about the Σ -imprimitive adjoint Selmer group, Proposition 6.1, which is the piece to put to an expert.

2. THE FRAMEWORK

2.1. Diamond–Flach–Guo. For a newform f of weight 2 and level N with rational coefficients, a prime $\lambda = \ell$ not dividing $2N$ ($k! = 2$), and a finite set Σ of primes, let $N^\Sigma = N \prod_{p \in \Sigma} p^{\delta_p}$ and let $f^\Sigma = \sum_{(n, \Sigma)=1} a_n q^n$ be the Σ -depleted eigenform, of level N^Σ . The integral cohomology of $X_0(N^\Sigma)$ carries an alternating pairing; its restriction to the rank-two lattice $M_{f^\Sigma, \lambda}$ (the f^Σ -isotypic part) has a discriminant, and [2, §1.7.3] define the *naive Σ -finite congruence ideal* $\eta_f^\Sigma \subset \mathbb{Z}_\ell$ by $\wedge^2 M_{f^\Sigma, \lambda} = \eta_f^\Sigma \cdot (\text{standard})$; it is an integral ideal.

Theorem 2.1 ([2, Proposition 1.4(c)]). *If $\bar{\rho}_f$ is irreducible then*

$$\eta_{f, \lambda}^\Sigma = \eta_{f, \lambda}^\emptyset \prod_{p \in \Sigma} L_p^{\text{nv}}(A_f, 1)^{-1}.$$

Only the primes with $\delta_p \geq 1$ contribute: adding an additive prime to Σ changes neither the level nor the ideal, adding a good prime multiplies the ideal by $(1 - \frac{\alpha}{\beta}q^{-1})(1 - q^{-1})(1 - \frac{\beta}{\alpha}q^{-1})$, Wiles’s level-raising factor, and adding a multiplicative prime multiplies it by $1 - q^{-2}$.

Theorem 2.2 ([2, Theorem 0.2 = 3.7]). *Let Σ contain the primes dividing N and let $\lambda \nmid 2N \prod_{p \in \Sigma} p$ be such that $\bar{\rho}_f|_{\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ is absolutely irreducible. Then the Σ -imprimitive adjoint Selmer group $H_\Sigma^1(G_{\mathbb{Q}}, A_{f,\lambda}/A_{f,\lambda})$ has length $v_\lambda(\eta_f^\Sigma)$.*

The same authors modify the integral structures in their Section 1.8 “to account for the congruences corresponding to Euler factors missing from $L^{\text{nv}}(A_f, s)$ ”, that is, at the exceptional primes, and prove the Bloch–Kato conjecture for A_f and $A_f(1)$ (their Theorem 0.1) for λ outside an explicit set S_f (primes dividing $2N$ and primes where the irreducibility over $\mathbb{Q}(\sqrt{\ell^*})$ fails).

2.2. Kim–Ota. In the formulation of [3, §2], which follows [2] and Dimitrov: under Assumption 2.1 ((FL): $k \leq \ell - 1$ and $\ell \nmid N$; (TW): $\bar{\rho}|_{\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ absolutely irreducible) and for Σ containing their exceptional set $P_{\bar{\rho}}$, the Σ -ramified deformation ring is the Hecke algebra T_Σ , a complete intersection, the Hecke module is free of rank two over it, and for every newform f in T_Σ , $\#\mathcal{O}/\eta_{f,\Sigma} = \#\text{Sel}_\Sigma(\mathbb{Q}, \text{ad}^0 f \otimes E/\mathcal{O})$ (their Theorem 2.9). Their Lemma 2.7 identifies the congruence ideal of the depleted form f^Σ at level N_f^Σ in the full Hecke algebra with $\eta_{f,\Sigma}$. Their main theorem (1.1) is the quantitative level lowering recalled in the note: for $N = N^+N^-$ with N^- square-free, $\ell \geq 5$, $\ell \nmid N$, (TW), and (6): $\bar{\rho}$ ramified at every $q \mid N^-$ with $q \equiv \pm 1 \pmod{\ell}$,

$$(2) \quad \text{ord}_\lambda \eta_f(N) = \text{ord}_\lambda \eta_f(N^+, N^-) + \sum_{q \mid N^-} t_f(q),$$

with $\eta_f(N)$ the congruence ideal in the full space and $t_f(q)$ the Tamagawa exponent (Definition 2.12 there), which is $t_\ell(E, q)$ for an elliptic curve. Remark 2.2 there, after [2, §1.7.1], says that one may assume the newforms to have minimal conductor among their twists when working with $\text{ad}^0 f$: this is the structural reason for the twist-symmetrised weights of the note.

2.3. Darmon–Diamond–Taylor. Three statements from [1] are used. (i) *The algebra of Lemma 4.17*: let $\mathbb{T}$ be a Hecke algebra acting on a lattice L free of finite rank over $\mathcal{O}$ with a perfect pairing for which $\mathbb{T}$ is self-adjoint, $\pi: \mathbb{T} \rightarrow \mathcal{O}$ an eigencharacter, $\wp = \ker \pi$, $I = \text{Ann}_{\mathbb{T}}(\wp)$, $\eta = \pi(I)$ the congruence ideal, and suppose $L[\wp]$ is free of rank d with Gram determinant Δ . Then $\eta^d \subseteq \Delta\mathcal{O}$, with equality if L is free over $\mathbb{T}$. The proof is three lines: $L/(L[\wp] \oplus L[I])$ is killed by η , generated by d elements, and of order $\#\mathcal{O}/\Delta$ by the duality $L/L[I] \simeq \text{Hom}(L[\wp], \mathcal{O})$. (ii) *The Greenberg–Wiles formula*, Theorem 2.18: for local conditions $\mathcal{L}$ on a finite module M , $\#H_{\mathcal{L}}^1(\mathbb{Q}, M)/\#H_{\mathcal{L}^*}^1(\mathbb{Q}, M^*) = \#H^0(\mathbb{Q}, M)/\#H^0(\mathbb{Q}, M^*) \cdot \prod_v \#\mathcal{L}_v/\#H^0(G_v, M)$. (iii) Corollary 2.43: the tangent space of R_Σ has dimension $\dim H_\Sigma^1(\mathbb{Q}, \text{ad}^0 \bar{\rho}(1)) + d_\ell + \sum_{p \in \Sigma - \{\ell\}} \dim H^0(\mathbb{Q}_p, \text{ad}^0 \bar{\rho}(1))$, “by the local Euler characteristic formula”. The terms $H^0(\mathbb{Q}_p, \text{ad}^0 \bar{\rho}(1))$ are the residual versions of the Euler terms of Definition 1.1: for the ℓ -adic module, $\#H^0(\mathbb{Q}_p, \text{ad}^0 \rho(1) \otimes K/\mathcal{O}) = \#\mathcal{O}/\det(1 - \text{Frob}_p | (\text{ad}^0 V)^{I_p}(1))$ when finite, and that determinant is the inverse Euler factor of A_f at $s = 1$. The Chapter 4 of [1] in which Theorem 4.20 (Hida’s formula) is proved assumes $\#\bar{\rho}(I_p) \mid \ell$ at every $p \neq \ell$ (their Section 3.3, hypothesis (e)); that excludes most of our additive types, which is why the proofs below use [2] instead.

3. THE LOCAL TERMS

3.1. Tame primes. Let $q \geq 5$, $q \neq \ell$. The inertia group acts on V through a finite cyclic quotient for potentially good reduction, and through a quadratic character times a unipotent for potentially multiplicative reduction. The invariants $(\text{ad}^0 V)^{I_q}$ and the Frobenius eigenvalue λ_q on them, in the arithmetic normalisation in which a good prime has eigenvalues $\alpha/\beta, 1, \beta/\alpha$ on $\text{ad}^0 V$, give $L_q(A_f, s)^{-1} = \prod (1 - \lambda q^{-s})$; the values of the table in Section 1 are obtained as follows, and were confirmed against PARI’s `lfunsympow` on every curve of the tables.

Multiplicative, I_n . The local component of f is an unramified twist of the Steinberg representation; $L_q(A_f, s) = (1 - q^{-1-s})^{-1}$, which is also the naive factor since $a_q^2 = 1$. So $e_\ell = 0$. I_n^* .

$V = V' \otimes \chi_q$ with V' Steinberg at q , $\text{ad}^0 V = \text{ad}^0 V'$, $a_q = 0$: $e_\ell = v_\ell(1 - q^{-2}) = v_\ell(q^2 - 1)$. II, II*, IV, IV*. The image of I_q in $\text{Aut } V$ is cyclic of order 3 or 6, generated by σ or $-\sigma$ with σ of order 3, eigenvalues $\zeta_3^{\pm 1}$. On $\text{ad}^0 V$ the sign disappears and the invariants form the line spanned by $\sqrt{-3} \in \mathbb{Z}_\ell[\zeta_3] \simeq \text{End}_{I_q}(V)$; Frobenius centralises σ if $q \equiv 1 \pmod{3}$ and inverts it if $q \equiv 2$, acting on the line by $\chi_{-3}(q)$. So $e_\ell = v_\ell(1 - \chi_{-3}(q)q^{-1}) = v_\ell(q - \chi_{-3}(q))$. III, III*. Image of order 4, eigenvalues $\pm i$; the invariant line of $\text{ad}^0 V$ is the diagonal element, Frobenius acts by $\chi_{-4}(q)$; $e_\ell = v_\ell(q - \chi_{-4}(q))$. I_0^* . $V = V' \otimes \chi_q$ with V' unramified, Frobenius eigenvalues α, β ; $\text{ad}^0 V$ unramified with eigenvalues $\alpha/\beta, 1, \beta/\alpha$, so $L_q(A_f, 1)^{-1} = (1 - \frac{\alpha}{\beta q})(1 - \frac{1}{q})(1 - \frac{\beta}{\alpha q}) = \frac{q-1}{q} \cdot \frac{(q+1)^2 - a_q(E')^2}{q^2}$.

3.2. Wild primes and the prime ℓ . At $q = 2, 3$ the Kodaira type does not determine $L_q(A_f, s)$, but PARI does, and Definition 1.1 applies verbatim; for instance a type IV at 2 has $P_2(X) = 1 + 2X$ and $e_3 = v_3(1 + \frac{1}{2}) = 1$, a type II at 2 has $P_2 = 1$ and $e_3 = 0$, and a type I_n^* at 2 has potentially good reduction in about half of the cases, with a cubic P_2 and $t = 0$ (this resolves the irregular behaviour of the prime 2 reported in the note). At $q = \ell$ the Euler factor is not the relevant quantity (the term at ℓ in Hida's formula is different), and the weights w_3 of Conjecture 1.2 are empirical.

4. TWIST MONOTONICITY

Theorem 4.1. *Let $E/\mathbb{Q}$ have conductor N , S a set of odd primes, $D = \prod_{q \in S} q$, χ_S the product of the quadratic characters of conductor $q \in S$, $E_S = E \otimes \chi_S$ of conductor N_S . If $q^2 \mid N$ for every $q \in S$ and $\text{lcm}(N_S, D^2) \mid N$, then $r_{E_S} \mid r_E$; if moreover $\text{lcm}(N, D^2) \mid N_S$, then $r_{E_S} = r_E$.*

Proof. Let $f = f_E$, $g = f_{E_S}$ and $\iota: h \mapsto h \otimes \chi_S = \sum \chi_S(n) a_n(h) q^n$, which maps $S_2(\Gamma_0(N_S))$ into $S_2(\Gamma_0(\text{lcm}(N_S, D^2))) \subset S_2(\Gamma_0(N))$, preserves integrality, and sends eigenforms to eigenforms with eigenvalues twisted by $\chi_S(p)$ at $p \nmid D$ and coefficients at multiples of D killed. Then $\iota(g) = f$: both are multiplicative, $a_p(f) = \chi_S(p) a_p(g)$ for $p \nmid D$ since the twist is unramified outside S , and $a_{q^k}(f) = 0$ for $q \in S$ since $q^2 \mid N$. An eigenform $h \neq g$ of level dividing N_S has an eigensystem different from that of g at infinitely many primes, so $\iota(h) \in f^\perp$. Hence a congruence $g \equiv h \pmod{r}$ with $h \in g^\perp$ integral gives $f \equiv \iota(h) \pmod{r}$ with $\iota(h) \in f^\perp$ integral, and $r \mid r_E$ by the cyclicity of the congruence module (Lemma 3.1 of the note). With $r = r_{E_S}$ this is the first claim; the second hypothesis makes the situation symmetric. $\square$

The hypotheses hold when (a) E has type I_n^* at each $q \in S$ ($N_S = N/D$), (b) type I_0^* at each $q \in S$ ($N_S = N/D^2$, E_S good at S), (c) type II, III, IV or a dual at each $q \in S$, $q \geq 5$ ($N_S = N$, equality). On the working set, for the 157,938 pairs (curve, prime of type I_n^*) and the 292,580 pairs (curve, prime of potentially good additive type), $v_\ell(\deg \varphi_E) \geq v_\ell(\deg \varphi_{E_q})$ holds for $\ell \in \{3, 5, 7, 11, 13\}$, $\ell^2 \nmid N$, with no exception, with equality in every pair of case (c) and strict inequality only in case (b), where the difference is the Euler term of Theorem 5.1 (script `step9_twist_degree.py`).

5. THE EULER TERMS OF THE TYPES I_0^* AND I_n^*

Theorem 5.1. *Let ℓ be an odd prime with $\ell \nmid 2N$ and $E[\ell]$ irreducible. Let S be a set of primes $q \geq 5$ at which E has type I_0^* or I_n^* , and $E_S = E \otimes \chi_S$. Then*

$$v_\ell(\deg \varphi_E) \geq v_\ell(r_{E_S}) + \sum_{q \in S} e_\ell(E, q),$$

with $e_\ell(E, q) = v_\ell(q - 1) + v_\ell((q + 1)^2 - a_q(E_S)^2)$ for type I_0^* and $v_\ell(q^2 - 1)$ for type I_n^* .

Proof. Write $f = f_E$, $g = f_{E_S}$ of level N_S , $D = \prod_{q \in S} q$, and $\Sigma = S$. By Lemma 2.4 of the note, E_S has good reduction at the I_0^* primes of S and multiplicative reduction at the I_n^* primes, so $\delta_q = 2$, resp. 1, for g at $q \in S$, and $N_S^\Sigma = N_S \prod_{q \in S} q^{\delta_q} = N$: the S -depleted form g^S lives at level N , and $\iota(g^S) = f$ with ι the twist map of Theorem 4.1 (its proof only used $a_{q^k}(f) = 0$ for $q \in S$ and ι on the prime-to- D coefficients, so it applies to g^S as well as to g).

Step 1: the twist map and the full Hecke algebra. Let $\mathbb{T}$ be the full Hecke algebra of level N and $\mathfrak{m}$ the maximal ideal of g^S , which contains U_q for $q \in S$. A normalised eigenform for $\mathbb{T}$ whose eigencharacter is congruent to that of g^S modulo $\mathfrak{m}$ has $U_q \equiv 0$ for $q \in S$. The q -stabilisations of g other than g^S have U_q -eigenvalue $a_q(g) = \pm 1$ (multiplicative case) or a root α, β of $X^2 - a_q(g)X + q$ (good case), which are ℓ -adic units since $\ell \neq q$; so they are not congruent to g^S modulo $\mathfrak{m}$, and the forms congruent to g^S are either new at every $q \in S$ with q^2 in their level, or old from newforms $h \neq g$ at primes outside S . For both kinds $\iota(h)$ is an eigenform of level dividing N with eigensystem different from that of f (at $p \nmid DN$ the eigensystems of h and g differ at infinitely many primes; at $q \in S$ both have $U_q = 0$), hence $\iota(h) \in f^\perp$. Therefore the argument of Theorem 4.1 gives: if $g^S \equiv h \pmod{\lambda^t}$ coefficientwise for an integral h in the full-Hecke-algebra orthogonal complement of g^S supported on the eigenforms above, then $f \equiv \iota(h) \pmod{\lambda^t}$ with $\iota(h) \in f^\perp$ integral. By the duality $S_2(\Gamma_0(N); \mathbb{Z}) \simeq \text{Hom}(\mathbb{T}, \mathbb{Z})$ and the cyclicity argument of Lemma 3.1 of the note (applied to the eigencharacter $\pi: \mathbb{T} \rightarrow \mathbb{Z}$ of g^S , which has $\mathbb{T}/\ker \pi = \mathbb{Z}$), the ℓ -part of the congruence number of g^S in $\mathbb{T}$, namely $v_\ell(\pi(\text{Ann}_{\mathbb{T}} \ker \pi)) =: v_\ell(\eta^{\text{Hecke}}(g^S))$, is attained by such h , and $v_\ell(r_E) \geq v_\ell(\eta^{\text{Hecke}}(g^S))$.

Step 2: Hecke versus cohomological congruence ideal. Apply the algebra of [1, Lemma 4.17] to $L = H_1(X_0(N), \mathbb{Z}) \otimes \mathcal{O}$ localised at $\mathfrak{m}$, with its intersection pairing made $\mathbb{T}$ -self-adjoint as in [1, §4.4], and to $\pi: L[\varphi]$ is the rank-two lattice $M_{g^S, \lambda}$ of [2, §1.7.3], whose Gram determinant is $(\eta_{g, \lambda}^S)^2$ up to a unit, the pairing being alternating. Hence $\eta^{\text{Hecke}}(g^S)^2 \subseteq (\eta_{g, \lambda}^S)^2$, that is, $v_\ell(\eta^{\text{Hecke}}(g^S)) \geq v_\ell(\eta_{g, \lambda}^S)$. (Freeness is not needed for this inequality.)

Step 3: the formula of Diamond, Flach and Guo. By Theorem 2.1, applied to g ($\bar{\rho}_g = \bar{\rho} \otimes \chi_S$ is irreducible, $\ell \nmid 2N_S$),

$$v_\lambda(\eta_{g, \lambda}^S) = v_\lambda(\eta_{g, \lambda}^\emptyset) + \sum_{q \in S} v_\ell(L_q^{\text{nv}}(A_g, 1)^{-1}) = v_\lambda(\eta_{g, \lambda}^\emptyset) + \sum_{q \in S} e_\ell(E, q),$$

the last equality because $A_g = A_f$ and, at $q \in S$, g has $\delta_q \geq 1$ so that its naive factor is the true one, which is the factor that f (with $a_q(f) = 0$) omits.

Step 4: the level N_S . At level N_S the form g is new, $\ell \nmid N_S$ and $\bar{\rho}_g$ is irreducible with Serre weight $2 \neq \ell$, so multiplicity one holds for the maximal ideal of g ([4, Theorem 3.5] and the references in its proof: the only possible exception there has Serre weight ℓ , not 2), hence the localised Hecke algebra is Gorenstein and $H_1(X_0(N_S), \mathcal{O})_{\mathfrak{m}}$ is free of rank two over it; the equality case of Lemma 4.17 then gives $v_\lambda(\eta_{g, \lambda}^\emptyset) = v_\ell(r_{E_S})$. Putting the steps together and using $v_\ell(\deg \varphi_E) = v_\ell(r_E)$ (as $\ell \nmid N$) proves the theorem. Without Step 4 one still has $v_\ell(\deg \varphi_E) \geq \sum_{q \in S} e_\ell(E, q)$, since $\eta_{g, \lambda}^\emptyset$ is an integral ideal. $\square$

Theorem 5.2. *Let $\ell \geq 5$, $\ell \nmid N$, with $\bar{\rho}_{E, \ell}$ absolutely irreducible on $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$. Let M be the set of primes q at which E has type I_n , or I_n^* with q odd, with $\ell \mid n$, and $M' \subset M$ the subset of primes $q \not\equiv \pm 1 \pmod{\ell}$. Then $v_\ell(r_{E_S}) \geq \sum_{q \in M'} t_\ell(E, q)$ for S the set of I_n^* primes in M' , and hence $v_\ell(\deg \varphi_E) \geq \sum_{q \in M'} t_\ell(E, q)$.*

Proof. E_S is multiplicative at every prime of M' with the same index, has conductor prime to ℓ , and $\bar{\rho}_{E_S} = \bar{\rho} \otimes \chi_S$ satisfies (TW). Apply (2) to f_{E_S} with $N^- = \prod_{q \in M'} q$: square-free, prime to N^+ , and (6) holds vacuously. Then use Theorem 4.1(a). $\square$

Corollary 5.3. *Under the hypotheses of Theorem 5.2, let S be the set of primes $q \geq 5$ of type I_0^* or I_n^* . Then*

$$v_\ell(\deg \varphi_E) \geq \sum_{q \in S} e_\ell(E, q) + \sum_{q \in M'} t_\ell(E, q) :$$

the Euler term and the Tamagawa term of a prime of type I_n^ add. For $\ell = 3$ and any set S of primes $q \geq 5$ of type I_n^* or I_0^* , Theorem 5.1 alone gives $v_3(\deg \varphi_E) \geq \sum_{q \in S} e_3(E, q)$, so a single prime of type I_n^* , with $3 \mid n$ or not, forces $3 \mid \deg \varphi_E$, and a prime of type I_0^* with $q \equiv 1 \pmod{3}$ does too.*

Proof. Theorem 5.1 for S gives $v_\ell(\deg \varphi_E) \geq v_\ell(r_{E_S}) + \sum_S e_\ell$; the twist E_S is multiplicative at S and Theorem 5.2 applied to E_S (whose I_n primes include M' with the same indices) gives $v_\ell(r_{E_S}) \geq \sum_{M'} t_\ell$, where for the I_n^* primes of M' no further twist is needed. $\square$

Remark 5.4. (1) Theorem 5.1 is the quantitative form of the twist-and-lower argument of the note, read in the other direction: the level is raised, from N_S to $N_S \prod q^{\delta_q} = N$, and the price is the level-raising Euler factor. (2) The only inputs are the formula of [2] for the naive congruence ideal, which needs $\bar{\rho}$ irreducible and $\ell \nmid 2N$, the algebraic inequality of [1, Lemma 4.17], and multiplicity one at the lower level. In particular $\ell = 3$ is allowed, and no Taylor–Wiles hypothesis enters. (3) Step 1 is where the argument would fail for the potentially good types II, IV, III: there the twist does not lower the level, f is its own depletion, and Theorem 2.1 gives no factor; the Euler factor sits inside the naive L -value, which is where Section 6 picks it up.

6. THE REMAINING TYPES: A SELMER-GROUP PROPOSITION

Let Σ be the set of primes dividing N and $M = \text{ad}^0 V \otimes K/\mathcal{O}$. Under the hypotheses of Theorem 2.2, $v_\ell(\eta_f^\Sigma) = \text{length } H_\Sigma^1(\mathbb{Q}, M)$, and by Theorem 2.1, $v_\ell(\eta_f^\Sigma) = v_\ell(r_E) + \sum_{p \parallel N} v_\ell(p^2 - 1)$ (multiplicity one at level N identifying $\eta_f^\emptyset$ with r_E). So the Euler-factor rule, for all types at once, is the statement

$$(3) \quad \text{length } H_\Sigma^1(\mathbb{Q}, M) \geq \sum_{p \in \Sigma, \delta_p=0} e_\ell(E, p) + \sum_{p \in \Sigma, \delta_p=1} (t_\ell(E, p) + v_\ell(p^2 - 1)).$$

Compare H_Σ^1 with the Bloch–Kato Selmer group $H_f^1(\mathbb{Q}, M)$ through the Greenberg–Wiles formula applied to the finite layers: $H_\Sigma^1 \supseteq H_f^1$, the local index at $p \in \Sigma$ between the unrestricted condition and the Bloch–Kato (unramified) condition is, by local Tate duality and the local Euler characteristic formula, $\#H^0(\mathbb{Q}_p, M^*) = \#H^0(\mathbb{Q}_p, \text{ad}^0 \rho(1) \otimes K/\mathcal{O})$ when the latter is finite, which is the ℓ -part of $L_p(A_f, 1)^{-1}$, i.e. $\ell^{e_\ell(E, p)}$ at an exceptional prime and $\ell^{v_\ell(p^2-1)+\dots}$ at a Steinberg prime; and the global correction is the ratio of the dual Selmer groups $\#H_{f*}^1(\mathbb{Q}, M^*)/\#H_{\Sigma*}^1(\mathbb{Q}, M^*) \geq 1$, with $M^* = \text{ad}^0 \rho(1) \otimes K/\mathcal{O}$ the module whose Bloch–Kato conjecture [2] also prove. Hence:

Proposition 6.1 (the statement to prove). *With $\ell \nmid 2N$ and (TW): the Euler-factor rule for the types II, IV, III and their duals is equivalent to*

$$\begin{aligned} & \text{length } H_f^1(\mathbb{Q}, \text{ad}^0 \rho \otimes K/\mathcal{O}) + \sum_{p \in \Sigma} \text{length } H^0(\mathbb{Q}_p, \text{ad}^0 \rho(1) \otimes K/\mathcal{O}) \\ & - \text{length } \frac{H_{f*}^1(\mathbb{Q}, \text{ad}^0 \rho(1) \otimes K/\mathcal{O})}{H_{\Sigma*}^1(\mathbb{Q}, \text{ad}^0 \rho(1) \otimes K/\mathcal{O})} \geq \text{right-hand side of (3)}, \end{aligned}$$

and the data of Section 7 say that the Bloch–Kato Selmer group supplies the Tamagawa exponents and the dual correction vanishes, except in the configuration of Section 7.3, where it equals one.

This is the precise form of the referees' question: the twist-symmetrised weights are explained by the twist invariance of $\text{ad}^0 \rho$, the exact local factors are the $H^0(\mathbb{Q}_p, \text{ad}^0 \rho(1) \otimes K/\mathcal{O})$ of Corollary 2.43 of [1] (whose residual version is the Taylor–Wiles count), and the one thing not in the literature as far as we can see is the vanishing of the dual correction, whose failure the type III anomaly exhibits. The refined integral structures of [2, §1.8], built precisely to account for the missing Euler factors at exceptional primes, are the natural place to settle it.

7. THE DATA

All computations are on the working set ($N \leq 300000$), hold-out ($300000 < N \leq 400000$), extension ($400000 < N \leq 500000$) and the 893 out-of-table curves of the note, with E optimal, non-CM, $E[\ell]$ irreducible; scripts `step9_*.py` (Kodaira-type formulas, $q \geq 5$) and `step10_*.py` (PARI's Euler factors at every bad prime) in the repository, outputs in `results/`.

7.1. Each term in isolation. For curves with exactly one additive prime $q \geq 5$, $q \neq \ell$, and no multiplicative prime with $\ell \mid n$, Table 1 gives the minimum of $v_\ell(\deg \varphi_E)$ by type and by $e_\ell(E, q)$. In every cell, for $\ell = 3, 5, 7$, the minimum equals e_ℓ and the proportion of curves with $v_\ell(\deg \varphi_E) \geq e_\ell$ is 1. For I_n^* the excess over e_ℓ has minimum exactly $v_\ell(n)$ in every cell. Multiplicative primes show no Euler term: for curves with a single multiplicative prime q with $\ell \mid n$ and no additive prime, $\min(v_\ell(\deg \varphi_E) - v_\ell(n)) = 0$ for every value of $v_\ell(q^2 - 1)$ up to 6.

type ($\ell = 3$)	$e = 1$		$e = 2$		$e = 3$		$e = 4$	
	min	n	min	n	min	n	min	n
II	1	8,327	2	821	3	59	4	2
II*	1	4,184	2	296	3	12	4	1
IV	1	4,290	2	296	3	12	4	1
IV*	1	8,221	2	821	3	59	4	2
III	1	1,271	2	76	3	9	–	–
III*	1	1,271	2	76	3	9	–	–
I_0^*	1	1,370	2	4,432	3	2,737	4	247
I_n^*	1	30,497	2	2,589	3	145	4	13

TABLE 1. Working set, $\ell = 3$: minimum of $v_3(\deg \varphi_E)$ (number of curves) over curves whose only additive prime is a single $q \geq 5$ of the given type with $e_3(E, q) = e$ and no multiplicative 3-contribution. The cells with $e = 0$ (III: 5,534 curves, I_0^* : 5,394) have minimum 0; for I_0^* the term runs up to $e = 6$ (minimum 6). The same holds at $\ell = 5$ and 7 (`results/step9_euler_15.txt`, 17).

7.2. The rule on whole ranges. Table 2 checks (1) with PARI's Euler factors at every bad prime $q \neq \ell$ and the terms t_ℓ , w_3 as defined, on the four ranges: no violation at any ℓ ; and it reports what happens when the type III terms are added (exceptions at $\ell = 3$ only) and when the weights at $q = \ell$ are dropped (none needed at $\ell \geq 5$). The equality rate at $\ell = 3$ is 57% on the working set, against 39% for the twist rule of the note and 48% for the version of (1) restricted to $q \geq 5$.

7.3. The type III anomaly. With the III terms included, all exceptions are at $\ell = 3$ and have one shape: a prime q of type III or III* whose Euler factor is nontrivial with Frobenius acting trivially on the invariant line ($q \equiv 1 \pmod{12}$ for $q \geq 5$; at $q = 2$ the factor $1 + 2X$), together with a prime $p \equiv -1 \pmod{3}$ with $t_3(E, p) \geq 1$ (a multiplicative prime, or one of type I_n^* with $3 \mid n$), and $v_3(\deg \varphi_E)$ falls short by exactly one. On the clean configurations of the working set (one such III prime, one such p , nothing else; 1,066 curves) the failure rate is 2%, it is 0 for $q \equiv 11 \pmod{12}$ (646 curves), and it rises with $v_3(q - 1)$ and with $v_3(p + 1)$ (to 12% for $p \equiv 8 \pmod{9}$);

set	ℓ	curves	(1) > 0	viol.	equality	viol. +III	no $q = \ell$: > 0 / viol. / equality
working set	3	1,233,729	861,074	0	56.8%	364	800,833 / 0 / 52.1%
	5	1,303,224	330,477	0	75.5%	0	330,477 / 0 / 75.5%
	7	1,307,753	173,488	0	84.1%	0	173,488 / 0 / 84.1%
	11	1,309,103	59,495	0	91.5%	0	59,495 / 0 / 91.5%
	13	1,308,945	35,635	0	92.4%	0	35,635 / 0 / 92.4%
hold-out	3	409,577	295,973	0	53.9%	68	277,265 / 0 / 49.3%
	5	428,251	114,860	0	74.0%	0	114,860 / 0 / 74.0%
	7	429,280	60,580	0	83.2%	0	60,580 / 0 / 83.2%
	11	429,502	21,097	0	90.6%	0	21,097 / 0 / 90.6%
	13	429,498	12,681	0	92.3%	0	12,681 / 0 / 92.3%
extension	3	348,227	255,507	0	53.2%	43	238,359 / 0 / 48.1%
	5	358,244	99,401	0	73.2%	0	99,401 / 0 / 73.2%
	7	358,717	52,766	0	82.7%	0	52,766 / 0 / 82.7%
	11	358,830	19,136	0	90.6%	0	19,136 / 0 / 90.6%
	13	358,816	11,484	0	91.9%	0	11,484 / 0 / 91.9%
out of table	3	893	827	0	51.6%	0	804 / 0 / 42.3%
	5	893	111	0	82.0%	0	111 / 0 / 82.0%
	7	893	52	0	90.4%	0	52 / 0 / 90.4%
	11	893	19	0	94.7%	0	19 / 0 / 94.7%
	13	893	12	0	100.0%	0	12 / 0 / 100.0%

TABLE 2. The Euler-factor rule with PARI's factors at all primes: curves with positive bound, violations and equality rate for (1); violations when the III terms are added at every prime (“+III”); violations and equality rate without the weights at $q = \ell$ (“no $q = \ell$ ”).

it does not depend on whether the multiplicative reduction at p is split. For every other type the Euler term adds to the Tamagawa terms with no exception, in particular when all the Tamagawa primes are $\equiv -1 \pmod{3}$ (14,000 curves). Type III is the only type whose residual inertia image mod 3 is semisimple and non-scalar; in the language of Proposition 6.1, the exceptions are the cases where the dual Selmer correction is 1.

7.4. The prime ℓ and the wild primes. For $\ell = 5, 7$ no weight is needed at $q = \ell$ or at $q = 2, 3$ beyond Definition 1.1: the minima over curves with a single wild additive prime are 0 except t_ℓ for potentially multiplicative reduction at 3. For $\ell = 3$ and $q = 3$ (potentially good) the minima are 1 for II, IV, III*, 2 for II*, IV* and 0 for III, I₀*, which are the weights w_3 ; at $q = 2$ Definition 1.1 with PARI's factor reproduces every minimum of the note (type IV at 2: $e_3 = 1$; type II at 2: 0; type I_n* at 2: $t_3 = v_3(-v_2(j))$ when potentially multiplicative, else 0, which explains why the index n of the symbol was not the right invariant).

8. COMPARISON TABLE

Table 3 sets the weights of the note against the exact local terms and records what is proved for each type.

9. CHANGES MADE TO THE NOTE (10 OCTOBER 2026)

- (1) The former Conjectures 1.1 and 1.3 are replaced by Conjecture 1.2 (Conjecture 1.2 of the note); the component-group and twist-symmetrised forms are its Proposition 1.3.

type at $q \geq 5$	weight in the note	local representation	exact local term	status
I_n	$v_\ell(n)$	Steinberg $\otimes$ unr.	$t = v_\ell(n), e = 0$	Kim–Ota, $\ell \geq 5$, (TW), $q \not\equiv \pm 1$ or $\bar{\rho}$ ramified; data: all ℓ , all q
I_n^*	$v_\ell(n)$	$\chi_q \otimes$ Steinberg	$v_\ell(n) + v_\ell(q^2 - 1)$	Theorem 5.1 (e , all odd ℓ), Theorem 5.2 (t), Corollary 5.3 (sum, $\ell \geq 5$)
I_0^*	0	$\chi_q \otimes$ unramified	$v_\ell(q-1) + v_\ell((q+1)^2 - a_q(E')^2)$	Theorem 5.1, all odd ℓ
IV, IV*	$[\ell = 3]$	tame, order 3	$v_\ell(q - \chi_{-3}(q))$	qualitative: Theorem 1.5(a) of the note; quantitative: Proposition 6.1, data
II, II*	$[\ell = 3]$	tame, order 6	$v_\ell(q - \chi_{-3}(q))$	same as IV*, IV by Theorem 4.1(c)
III, III*	0	tame, order 4	$v_\ell(q - \chi_{-4}(q))$	attained in isolation; not additive in the configuration of Section 7.3
2, 3	empirical	wild	PARI's factor; t via $v_q(j)$	data, no exception
ℓ	empirical		w_3	data

TABLE 3. Weights of the note, exact local terms and their status, by Kodaira type.

- (2) Theorems 4.1, 5.1, 5.2 and Corollary 5.3 are Theorems 4.3, 4.4, 4.5 and Corollary 4.6 of the note, summarised as its Theorem 1.6; the former Theorem 1.4 (the qualitative statements at $\ell = 3$) is its Theorem 1.5.
- (3) Section 6 is Section 5 of the note, with Proposition 6.1 stated there as Conjecture 5.1 (Selmer form of the rule); the introduction presents the modular degree as governed by the naive adjoint L -value, the local terms as the omitted Euler factors, and the type III anomaly as the test case. This is the question for Kim and Ota, who have exactly the tools (their Theorem 2.13 and Proposition 2.15).
- (4) The equality rates of Table 2 are reported in Section 6.3 of the note as the measure of sharpness.

Statement on the use of an AI system. This memorandum, like the note it accompanies, was produced with an AI system (Claude, Anthropic) at every stage, under the author's direction and in discussion with the author: the literature search, the formulation of the Euler-factor rule, the computations with PARI's Euler factors, the proofs and the text. The work would not have been done without it. The author takes full responsibility for all statements, proofs, computations and errors; the citations that an expert should check are named in Remark 5.4.

REFERENCES

- [1] H. Darmon, F. Diamond, R. Taylor, Fermat's last theorem, in: Current developments in mathematics, 1995, Int. Press, 1994, 1–154 (theorem numbers as in the revised version at <https://www.math.mcgill.ca/darmon/pub/Articles/Expository/05.DDT/paper.pdf>).
- [2] F. Diamond, M. Flach, L. Guo, The Tamagawa number conjecture of adjoint motives of modular forms, Ann. Sci. École Norm. Sup. 37 (2004), 663–727.
- [3] C.-H. Kim, K. Ota, On the quantitative variation of congruence ideals and integral periods of modular forms, Res. Math. Sci. 10 (2023), Paper No. 22 (arXiv:1905.02926).
- [4] K. A. Ribet, W. A. Stein, Lectures on Serre's conjectures, in: Arithmetic algebraic geometry (Park City, 1999), IAS/Park City Math. Ser. 9, Amer. Math. Soc., 2001, 143–232.

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